VisionGuard
AI-Powered Driver Drowsiness Monitoring

A modular prototype that turns eye-state and yawning evidence into transparent, rule-based drowsiness warning candidates — in real time and from uploaded video.

Deep Learning Group Project · UTS · Live System Demo

Drowsiness is a multi-cue, time-based signal — not a single snapshot

01 Drowsy driving is safety-critical, yet a single frame rarely proves fatigue on its own.

02 The real cues live in the eyes and the mouth, and they only become convincing over several seconds.

03 A single end-to-end “drowsy / not-drowsy” box hides its reasoning and tends to break on unseen drivers.

04 Cameras fail: a turned head or a lost face must never be read as drowsiness.

05 A driver-assist tool should surface explainable evidence, not deliver a medical verdict.

> Split the problem into specialist visual signals, then fuse them over time with transparent rules.
>
> VISIONGUARD DESIGN PRINCIPLE

2	5	0
Specialist CNN models	Rule-based warning states	Black-box verdicts

Scope: VisionGuard outputs rule-based warning candidates and signal-quality flags — not a final system-level or medical drowsiness judgment.

SYSTEM ARCHITECTURE

End-to-End Flow

1 · User Interface Layer
Live Monitor · Video Upload Analysis · History · Insights

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2 · Input & Pre-processing Layer
Webcam or uploaded video · input validation · frame sampling · face detection · signal-quality check · eye ROI + mouth ROI extraction

▼

3A · Eye Specialist CNN
MobileNetV2 on MRL Eye crops — classifies eye state open / closed

Output → `p_eye_closed`

3B · Mouth / Yawn Specialist CNN
ResNet18 on YawDD / YawDD+ mouth crops — classifies no-yawn / yawn

Output → `p_yawn`

▼

4 · Temporal Rule-based Fusion Layer
Combines `p_eye_closed`, `p_yawn`, temporal evidence, and signal-quality flags

▼

Fusion output · Warning-Candidate States
normal · **eye_warning** · **mouth_warning** · **high_confidence_drowsiness** · **signal_unreliable**

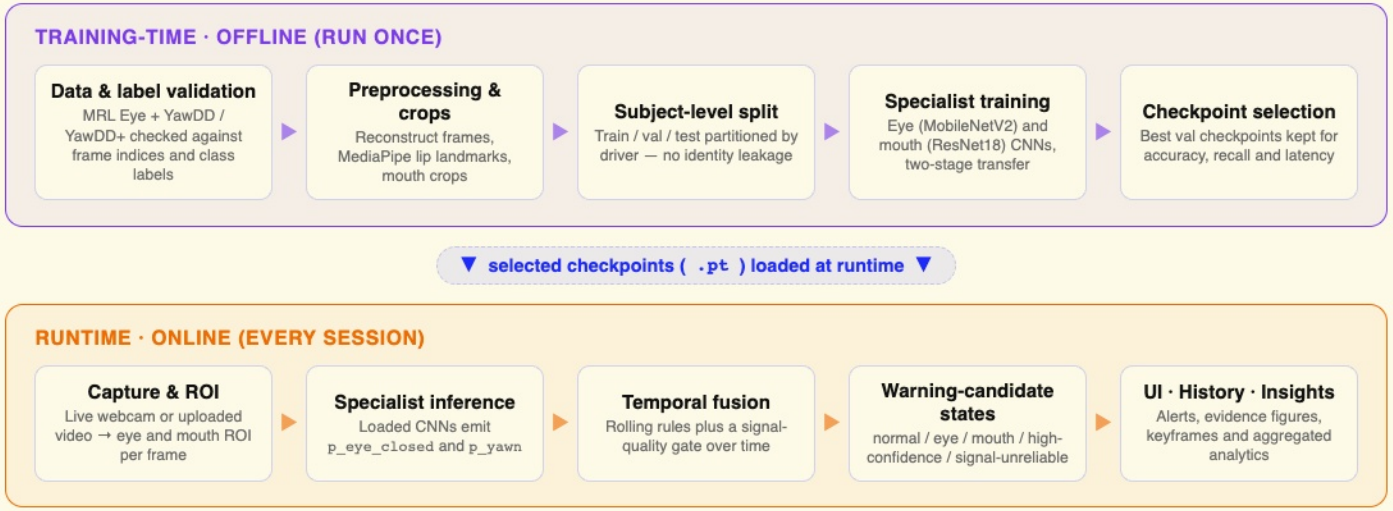
▼

5 · Result Storage & Reporting Layer
Analysis summaries · alert intervals · evidence figures · keyframes · history records · insight summaries

PROJECT WORKFLOW

Training-time → Runtime

From raw datasets to a live, explainable system



DATASET STRATEGY

Complementary Data

Two complementary datasets, each split by subject

84,898 images
MRL Eye open / closed corpus

Near-balanced: 41,946 closed vs 42,952 open.

64,202 mouth crops
YawDD + YawDD+ Dash yawning

99.73% crop success via MediaPipe lip landmarks.

37 subjects
MRL Eye — eye specialist

Split 25 / 6 / 6 across train · val · test.

29 drivers
YawDD Dash — mouth specialist

Split 20 / 4 / 5 across train · val · test.

100% subject-level
No driver appears in two splits

Prevents identity and frame leakage across splits.

2 label sets
closed / open · no_yawn / yawn

Specialist labels — not end-to-end drowsiness labels.

PREPROCESSING PIPELINE

Raw Video to Crops

Training data prep — the YawDD mouth pipeline

1

Reconstruct frames

YawDD+ frame index and class labels rebuild labeled full frames from the YawDD raw Dash videos.

2

Mouth crops

MediaPipe Face Mesh / FaceLandmarker lip landmarks crop the mouth region (99.73% success).

3

Haar fallback

OpenCV Haar lower-face crop when MediaPipe fails; failed crops are dropped and never used for training.

4

Subject split

Partition by driver so train, val, and test never share an identity.

MRL Eye already provides cropped eye images, so the eye specialist needs no frame reconstruction. Runtime is separate: from live webcam or uploaded video, both eye and mouth ROI are extracted per frame.

SPECIALIST MODELS

Two CNNs + Fusion

Each component owns one visible behavior

98.6%

MRL Eye · MobileNetV2 → `p_eye_closed`

Eye open / closed specialist, selected as the runtime eye model for its accuracy and latency balance.

— Macro F1 98.6% on held-out subjects
— Closed-eye recall 98.5%
— Decision threshold p ≥ 0.50

✓ Selected runtime eye model

99.4%

YawDD Mouth · ResNet18 → `p_yawn`

Mouth / yawn specialist trained on subject-split Dash mouth crops.

— Yawn F1 97.2%
— Yawn recall 97.9%
— Best of three CNN baselines

✓ Selected runtime mouth model

Rules

Temporal fusion → warning states

No trained fusion model — a transparent rule layer combines both signals over time with a quality gate.

— Five warning-candidate states
— PERCLOS-like proxy from rolling `p_eye_closed`
— Quality gate for lost-face frames

✓ Fully explainable

TRAINING PROTOCOL

How the CNN specialists were trained

Two-stage transfer learning on ImageNet backbones

Backbones & classifier head

- ImageNet-pretrained ResNet18, MobileNetV2, EfficientNet-B0
- ImageNet head replaced with a new 2-class `nn.Linear` head
- A standard PyTorch layer — not a custom or third-party head

Two-stage transfer

- Stage 1: freeze the backbone, train the new classifier head
- Stage 2: unfreeze the full backbone for fine-tuning

Optimization

- Loss: weighted `CrossEntropyLoss`
- Optimizer: Adam (mouth / yawn), AdamW (MRL eye)

Inputs & split

- 224×224 RGB with ImageNet normalization
- Subject-level train / val / test split — no identity leakage

MODEL EVIDENCE RESULTS

Benchmark

Specialist test accuracy across three CNN baselines

Subject-level test accuracy; bars scaled to a 97.5–100% window for contrast. Metrics are within-task specialist results — mouth and eye accuracies are not one unified drowsiness benchmark, and not final system-level drowsiness accuracy.

TEMPORAL DECISION FUSION

Evidence + Quality + Persistence

How warning states emerge from evidence over time

Eye evidence over time

- PERCLOS-like proxy from rolling `p_eye_closed`
- Eye warning when rolling mean ≥ 0.60
- Enter / exit hysteresis resists flicker

Mouth / yawn evidence

- `p_yawn` per mouth crop, thresholded to yawn events
- Recent-yawn context: 4 s live, 8 s upload
- A yawn alone is a candidate, not a verdict

Signal-quality gate

- Track the no-face / lost-ROI ratio
- Flag `signal_unreliable` when faces are missing
- Tracking failure is not drowsiness evidence

Persistence rules

- Require consecutive frames before alerting
- Debounce and cooldown on each alert kind
- High confidence needs sustained eye + recent yawn

Five candidate states

- normal · eye-warning · mouth-warning
- high-confidence drowsiness candidate
- signal unreliable

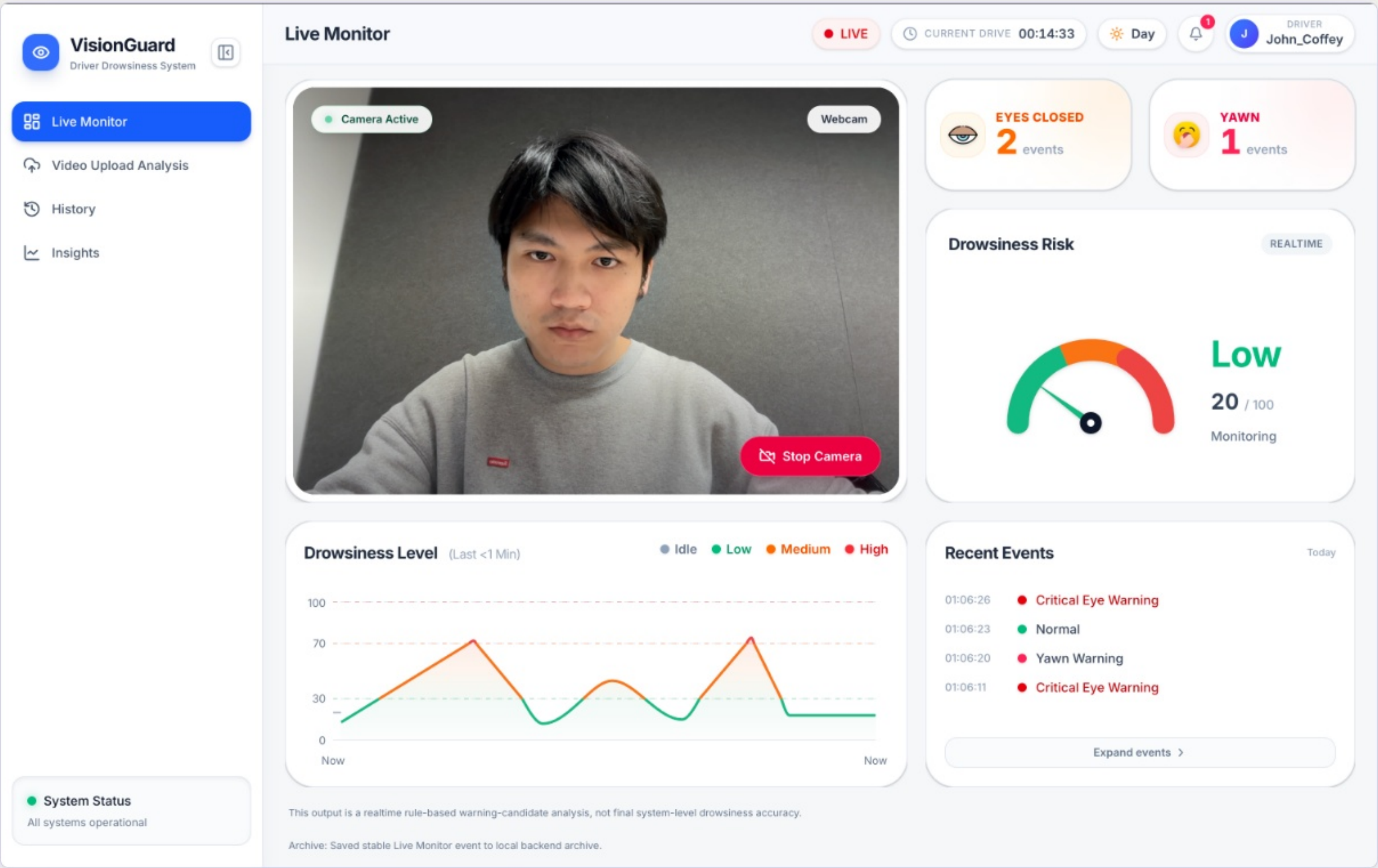
Demonstrated on a controlled clip

- 103 sampled frames, F5 tiered rule
- 18 eye-warning, 30 mouth-warning frames
- 6 high-confidence frames where both overlap
- Controlled-demo warning-candidate output — not final accuracy

WORKING SYSTEM INTERFACE

Live Product

Live Monitor | Video Upload Analysis | History | Insights



Live VisionGuard prototype · click a tab to switch views

From Evidence to Awareness

Live demo in three steps — start the camera and trigger an eye-closure warning, upload a clip to review fused intervals and keyframes, then open History & Insights to see the trail.

Start the Live Demo

VisionGuard · Rule-based warning candidates, not a final drowsiness verdict · Thank you